Cristopher Garduno Luna

Seattle, WA | Personal Website | Google Scholar

SUMMARY

Empirical researcher and data scientist with 10+ years of experience across academic laboratories and industry, working at the intersection of human perception, human movement, and wearable sensor data. I design human subject studies and build predictive models that integrate sensor, physiological, biomechanical, and survey data. I work with engineering, design, and product teams to turn evidence into decisions.

PROFESSIONAL EXPERIENCE

Brooks Sports, Inc., Seattle, WA

Run Behavior Researcher II (Mar 2026 to Present)

- Lead research that integrates data across disciplines to understand the runner and guide product decisions.
- Work across behavior, physiology, biomechanics, and data science to answer research questions quickly, using AI to build tools for data collection, processing, and analysis as needs arise.

Run Perception Researcher II (Aug 2025 to Mar 2026)

- Develop performance metrics that bridge objective device telemetry with subjective feedback to evaluate product experience.
- Serve on the company AI Working Group, evaluating responsible AI tool adoption and leading technical workshops for research and engineering staff.

Run Perception Researcher I (Sep 2022 to Aug 2025)

- Developed and validated models predicting biomechanical load from smartwatch sensor data, published in *Footwear Science* (2025).
- Ran in-field behavioral and sensor studies at ultra-endurance events, studying movement and perception under acute fatigue.
- Combined data from wearables with structured survey data and presented findings to product, design, and engineering teams.

Product Fit & Wear Test Analyst (Jun 2021 to Sep 2022)

- Ran evaluative testing programs across diverse participant cohorts, assessing fit, comfort, wear, and durability.
- Replaced manual feedback compilation with automated Python statistical scripts, speeding analysis cycles while preserving data validity.
- Improved survey workflows to raise the quality of feedback collected.

ACADEMIC RESEARCH

Graduate Researcher, Attention Lab & Action Lab, UC Santa Barbara (2017 to 2020). Visual perception and motor behavior under exertion; VR and AR experimental platforms.

Undergraduate Researcher, Hsiao Lab, UCLA (2016 to 2017). Computational and experimental neurotransmitter signaling research.

Research Assistant, Neurovascular Imaging Research Core, UCLA (2014 to 2016). Machine learning for medical image registration and vessel segmentation.

EDUCATION

M.A., Dynamical Neuroscience, University of California, Santa Barbara

B.S., Neuroscience, University of California, Los Angeles

PUBLICATIONS

- Modelling peak vertical ground reaction force from smart watch metrics reveals link between higher forces and softer cushioning perception (2025). *Footwear Science*.
- Weakly-Supervised Convolutional Neural Networks for Vessel Segmentation (2022). *IEEE/CVF WACV*.
- Feasibility of Virtual and Augmented Reality Devices as Psychology Research Tools (2020). University of California, Santa Barbara.
- Intestinal Serotonin and Fluoxetine Exposure Modulate Bacterial Colonization in the Gut (2019). *Nature Microbiology*.

SKILLS

- **Human movement & wearable sensing:** IMU and wearable time series analysis, sensor fusion, biomechanical modeling, field and longitudinal studies.
- **Data science & machine learning:** Python (Pandas, NumPy, SciPy, scikit-learn, PyTorch), SQL, predictive modeling, time series classification, statistical inference.
- **Research design & survey analytics:** human subject study design, psychophysics, psychometrics, survey design, multi-cohort studies.
- **Technical enablement:** data pipelines, analysis automation, internal tool prototyping, workshop facilitation.

Independent project: Sleep staging model. A PyTorch sequence classifier using sparse attention to label sleep stages from physiological time series.